

Unit 3 Test Study Guide ITERATION 2

Dependency Relationships | Graphs and Multiple Representations

Name: _____

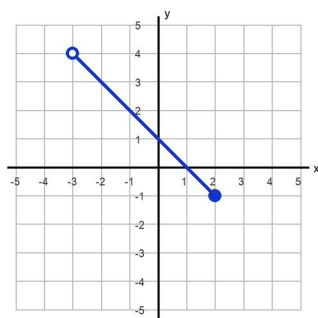
Period: _____ Due: _____

How to use this guide: This iteration adds graph-heavy practice to match the Unit 3 test: reading values from graphs, using domain/range with endpoints, comparing discrete and continuous representations, and interpreting linear graphs in context.

Multiple-Choice Practice

1A. Domain from a Graph | Version A

What is the domain of the graphed relation?

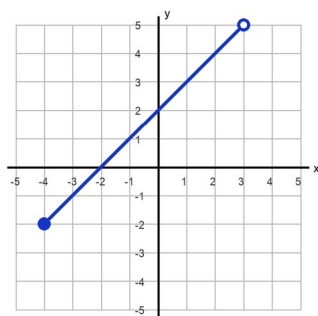


- A. $(-3, 2)$
- B. $[-3, 2]$
- C. $(-3, 2]$
- D. $[-1, 4]$

Work / thinking

1B. Range from a Graph | Version B

What is the range of the graphed relation?

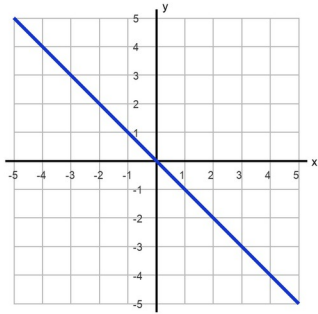


- A. $[-2, 5]$
- B. $[-2, 5)$
- C. $(-4, 3)$
- D. $(-2, 5]$

Work / thinking

2A. Reading Function Values from a Graph | Version A

The graph of $y = f(x)$ is shown. What is $f(1)$?

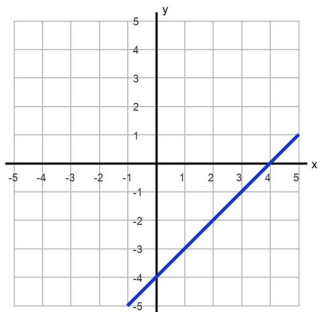


- A. -2
- B. -1
- C. 0
- D. 1

Work / thinking

2B. Reading Inputs from a Graph | Version B

The graph of $y = g(x)$ is shown. For what value of x is $g(x) = -2$?

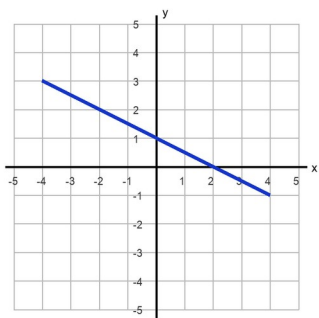


- A. $x = 1$
- B. $x = 2$
- C. $x = 3$
- D. $x = 4$

Work / thinking

3A. Slope from a Graph | Version A

Determine the slope of the graphed line.

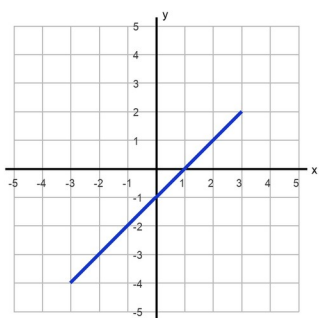


- A. 2
- B. -2
- C. $\frac{1}{2}$
- D. $-\frac{1}{2}$

Work / thinking

3B. Slope from a Graph | Version B

Determine the slope of the graphed line.

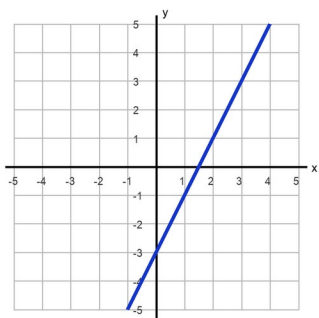


- A. 1
- B. -1
- C. 2
- D. $\frac{1}{2}$

Work / thinking

4A. Equation from a Graph | Version A

Which equation matches the graphed line?

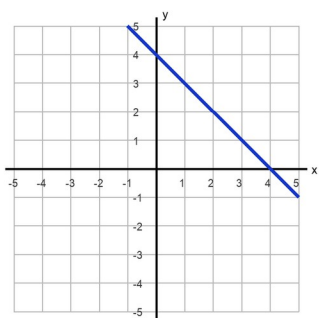


- A. $y = 2x - 3$
- B. $y = -2x - 3$
- C. $y = 2x + 3$
- D. $y = x - 3$

Work / thinking

4B. Equation from a Graph | Version B

Which equation matches the graphed line?



- A. $y = x + 4$
- B. $y = -x + 4$
- C. $y = -4x + 1$
- D. $y = x - 4$

Work / thinking

5A. Functions Across Representations | Version A

Which relation is NOT a function? A. $\{(0, 1), (1, 3), (2, 5)\}$ B. $\{(-2, 4), (1, 4), (3, 4)\}$ C. $\{(2, 6), (2, -1), (5, 0)\}$ D. $\{(-5, 0), (2, -1), (4, 3)\}$	Work / thinking _____ _____ _____
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5B. Discrete and Continuous | Version B

Which situation is continuous? A. number of books on a shelf B. number of students absent C. amount of water in a bottle D. number of goals scored	Work / thinking _____ _____ _____
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6A. Equation from a Table | Version A

For x : 0, 1, 2, 3 and y : 2, 7, 12, 17, which rule fits? A. $y = 2x + 5$ B. $y = 5x + 2$ C. $y = 7x - 5$ D. $y = 3x + 2$	Work / thinking _____ _____ _____
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6B. Function Notation | Version B

Given $h(x) = -3x + 8$, what is $h(4)$? A. -4 B. 4 C. 11 D. 20	Work / thinking _____ _____ _____
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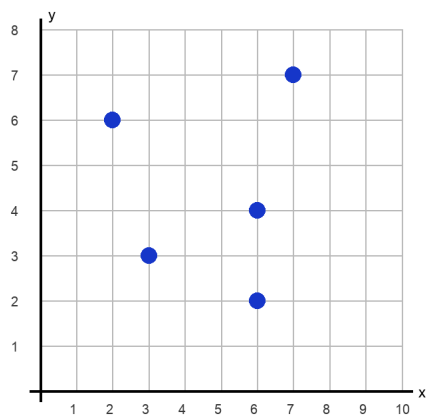
Free-Response Practice

Show clear work. For graph problems, mark the points or interval you used before writing the answer.

7-A. Graph Representation Analysis | Version A

Use the discrete graph.

- Is it discrete or continuous?
- State the domain.
- State the range.
- Is it a function?



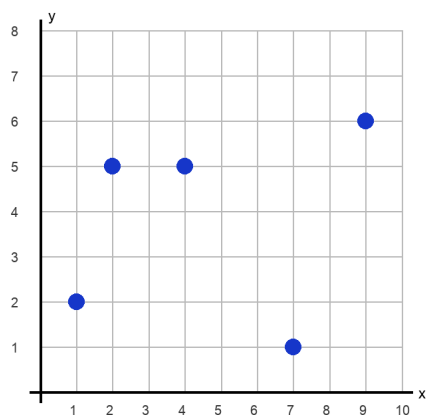
Answer: _____

Work / thinking

7-B. Graph Representation Analysis | Version B

Use the discrete graph.

- Is it discrete or continuous?
- State the domain.
- State the range.
- Is it a function?



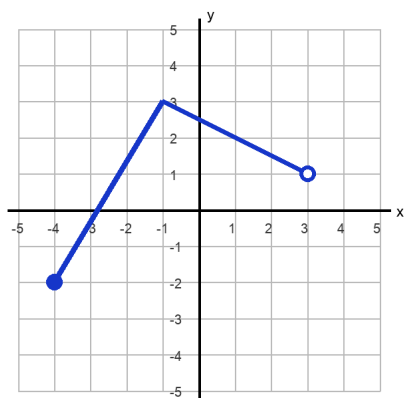
Answer: _____

Work / thinking

8-A. Continuous Graph Analysis | Version A

Use the graphed relation.

- Is it discrete or continuous?
- State the domain.
- State the range.
- Is it a function?



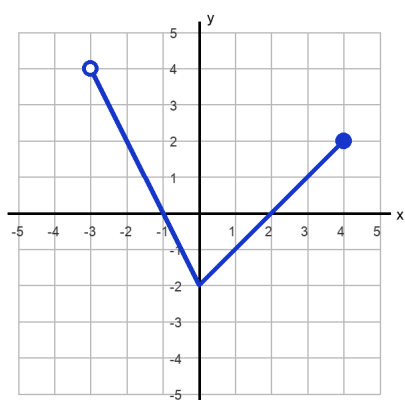
Answer: _____

Work / thinking

8-B. Continuous Graph Analysis | Version B

Use the graphed relation.

- Is it discrete or continuous?
- State the domain.
- State the range.
- Is it a function?



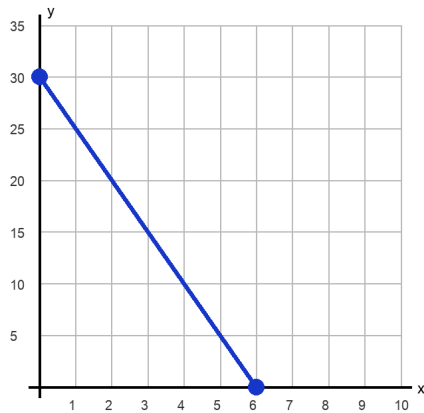
Answer: _____

Work / thinking

9-A. Context Graph Analysis | Version A

A tank starts leaking water. Use $V(t)$ from the graph.

- Find the slope and explain it.
- Write an equation.
- Find $V(0)$.
- Find t when $V(t) = 10$.
- State the domain and range.



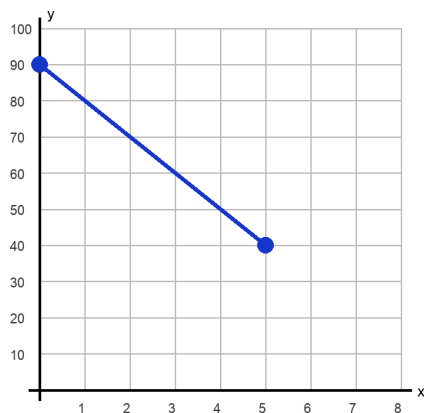
Answer: _____

Work / thinking

9-B. Context Graph Analysis | Version B

A phone battery drains during game play. Use $B(t)$ from the graph.

- Find the slope and explain it.
- Write an equation.
- Find $B(0)$.
- Find t when $B(t) = 60$.
- State the domain and range.



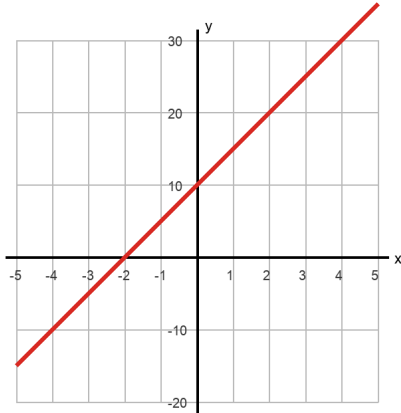
Answer: _____

Work / thinking

10-A. Linear Graph Features | Version A

Use the red graph of $y = f(x)$.

- Find $f(-4)$.
- Find x if $f(x) = 0$.
- Find the slope.
- Write the equation.



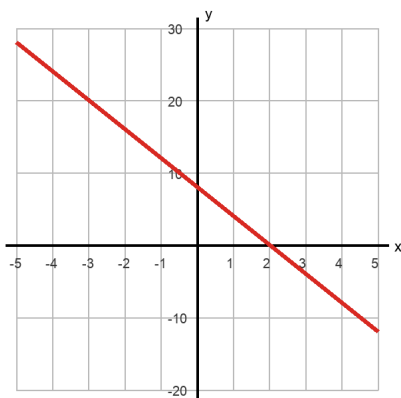
Answer: _____

Work / thinking

10-B. Linear Graph Features | Version B

Use the red graph of $y = g(x)$.

- Find $g(4)$.
- Find x if $g(x) = 0$.
- Find the slope.
- Write the equation.



Answer: _____

Work / thinking

11-A. Slope Formula | Version A

Find the slope through $(-1, 1)$ and $(2, -5)$. Show substitution. Answer: _____	Work / thinking
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11-B. Slope Formula | Version B

Find the slope through $(-3, -4)$ and $(5, 8)$. Show substitution. Answer: _____	Work / thinking
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12-A. Table to Linear Rule | Version A

For the table x: 0, 1, 2, 3, 4 and $f(x)$: 1, 4, 7, 10, 13: a. Is it linear? Why? b. Find $f(2)$. c. Find x if $f(x) = 10$. d. Write the rule. Answer: _____	Work / thinking
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12-B. Table to Linear Rule | Version B

For the table x: 0, 1, 2, 3, 4 and $g(x)$: 12, 10, 8, 6, 4: a. Is it linear? Why? b. Find $g(2)$. c. Find x if $g(x) = 6$. d. Write the rule. Answer: _____	Work / thinking
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Student Self-Check Answer Bank

Important: This is not a worked solution key. It is a self-check bank. If your answer is not here, go back and check the graph, endpoint notation, slope, or function setup.

Multiple-Choice Practice

Solve first, then check. Cross off each answer only after your work supports it.

C. $(-3, 2]$	A. 1
B. $[-2, 5)$	A. $y = 2x - 3$
B. $y = -x + 4$	C. amount of water in a bottle
B. $x = 2$	B. $y = 5x + 2$
C. $\{(2, 6), (2, -1), (5, 0)\}$	B. -1
D. $-1/2$	A. -4

Free-Response Practice

Solve first, then check. Cross off each answer only after your work supports it.

a. -8; b. $x = 2$; c. -4; d. $g(x) = -4x + 8$	a. -10 percent/hour; b. $B(t) = -10t + 90$; c. 90%; d. $t = 3$; e. domain $[0, 5]$, range $[40, 90]$
a. discrete; b. $\{2, 3, 6, 7\}$; c. $\{2, 3, 4, 6, 7\}$; d. not a function	a. -5 liters/min; b. $V(t) = -5t + 30$; c. 30 liters; d. $t = 4$; e. domain $[0, 6]$, range $[0, 30]$
a. continuous; b. $[-4, 3)$; c. $[-2, 3]$; d. function	$m = 3/2$
a. -10; b. $x = -2$; c. 5; d. $f(x) = 5x + 10$	$m = -2$
a. yes, constant difference 3; b. 7; c. $x = 3$; d. $f(x) = 3x + 1$	a. continuous; b. $(-3, 4]$; c. $[-2, 4)$; d. function
a. discrete; b. $\{1, 2, 4, 7, 9\}$; c. $\{1, 2, 5, 6\}$; d. function	a. yes, constant difference -2; b. 8; c. $x = 3$; d. $g(x) = -2x + 12$

Final Check Before the Test

- ☐ I can read domain and range from graphs using correct endpoint notation.
- ☐ I can find $f(x)$ and solve for x by reading a graph.
- ☐ I can determine slope and write a linear equation from a graph.
- ☐ I can decide whether a relation is a function across different representations.
- ☐ I can interpret slope, intercepts, domain, and range in context.